

```
i = 1      i
j = 1      j
```

---

The value at index  $j$  (which is 7) is not less than or equal to the pivot value (which is 4), so don't swap the values. Remember,  $j$  always increases, but  $i$  increases only after a swap is performed—so  $i$  is always either at or to the left of  $j$ . The state of the variables now looks like this:

---

```
items:    [0, 7, 6, 3, 1, 2, 5, 4]
indices:  0  1  2  3  4  5  6  7
           ^
i = 1      i  ^
j = 2      j
```

---

The value at index  $j$  (which is 6) is not less than or equal to the pivot value (which is 4), so don't swap the values. The state of the variables now looks like this:

---

```
items:    [0, 7, 6, 3, 1, 2, 5, 4]
indices:  0  1  2  3  4  5  6  7
           ^
i = 1      i      ^
j = 3      j
```

---

The value at index  $j$  (which is 3) is less than or equal to the pivot value (which is 4), so swap the values at  $i$  and  $j$ . The 7 and 3 swap positions. Also, increase  $i$  so that it moves to the right. The state of the variables now looks like this:

---

```
items:    [0, 3, 6, 7, 1, 2, 5, 4]
indices:  0  1  2  3  4  5  6  7
           ^
i = 2      i      ^
j = 4      j
```

---

The value at index  $j$  (which is 1) is less than or equal to the pivot value (which is 4), so swap the values at  $i$  and  $j$ . The 6 and 1 swap positions. Also, increase  $i$  so that it moves to the right. The state of the variables now looks like this: